

Qiyuan Shi

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RESEARCH INTERESTS

Controllable and accountable large language models: activation steering, synthetic training data, data-use auditing and membership inference, and LLM evaluation and safety.

EDUCATION

Rutgers University–New Brunswick Sep 2026 – present
PhD in Computer Science · Advisor: Prof. Ruixiang (Ryan) Tang New Brunswick, NJ

University of Science and Technology of China (USTC) Sep 2022 – Jul 2026
B.Eng. in Computer Science, School of the Gifted Young Hefei, China

- GPA: 85.53/100. Selected coursework: Fundamentals of Artificial Intelligence (98), Foundations of Algorithms (96), Mathematical Analysis (96), Introduction to Deep Learning (94), Linear Algebra (92), Probability and Mathematical Statistics (90).

PUBLICATIONS

Zheng Zhang, Yifan Wu, Yuxiang Zhou, Qi Liu, Qingyang Mao, Ning Li, Weibo Gao, **Qiyuan Shi**, Zhenya Huang, Baosheng Yu. **Beyond Single-Attribute Auditing: A Configurable Multi-Attribute Benchmark for Intersectional Fairness in LLMs.** *EMNLP 2026* (Main Conference).

MANUSCRIPTS UNDER REVIEW

UpdateSteer: Set-Conditioned Activation Control for Synthetic Training Data. Under review, 2026.

DAudit: Data-use Auditing of Large Language Models. Under review, 2026.

Membership Attribution against Large Language Models. Under review, 2026.

RESEARCH EXPERIENCE

UpdateSteer: Set-Conditioned Activation Control for Synthetic Data 2026 – present
PhD research, Rutgers University

- Jointly selects activation interventions for a set of synthetic examples, using learner gradients to represent each example’s update signal and set-level statistical matching to make examples complementary.
- Evaluated on toxicity detection, reward modeling, verifiable instruction following, and prompt-injection detection.

DAudit: Data-use Auditing of Large Language Models Jul 2025 – Jan 2026
Research Assistant, Johns Hopkins University · Advisor: Prof. Yinzhi Cao

- Black-box auditing of whether specific data were used in post-training, under label-only and logits access, with formal false-discovery-rate guarantees; supports copyright compliance and “Do-Not-Train” verification.
- Proposed the DualBank alignment scheme and built the end-to-end pipeline: data marking, per-sample Shadow-LoRA training, query scheduling, Δ -score computation, and sequential decision thresholds.
- Ran ablations on paraphrase/noise level, bank size, LoRA rank, target layers, and constraint strength, plus runtime and statistical-power analyses; co-authored the methods section.

Membership Attribution against Large Language Models

Sep 2025 – Feb 2026

Research Assistant, Johns Hopkins University · Advisor: Prof. Yinzhi Cao

- Moved beyond binary membership inference to attributing *which* training stage used a sample, under low false-positive constraints.
- Built attribution signals from cross-stage output-probability differences and trained a Transformer-based attack model; adapted it to restricted-logit access via probability recovery and entropy features.
- Reproduced Blind Attribution and six calibrated MI baselines; set up PT-PT (Pythia snapshots), FT-FT (FineWeb/UltraFeedback with Llama 3/Qwen 3), and PT-FT (WikiMIA with Llama 2/GPT-J/Pythia/OPT) settings.

FairRO: Fairness without Demographics from a Robust Optimization Perspective

Sep 2024 – 2026

Research Assistant, USTC · Advisor: Prof. Qi Liu

- Controls worst-group risk and exposure when only aggregate group proportions are known. Co-designed the ambiguity set and min-max training objective.
- Implemented the trainer and a pluggable re-ranking module; ran ablations on proportion misspecification, sample size, loss families, and constraint strength; prepared reproducible scripts and documentation.

Multi-Attribute Benchmark for Intersectional Fairness in LLMs (EMNLP 2026)

Jul 2024 – Jan 2025

Research Assistant, USTC · Advisor: Prof. Qi Liu

- Implemented baselines and a unified prompting pipeline; ran multi-model, multi-dataset experiments with uncertainty analysis; drafted the results and discussion.

TEACHING

Teaching Assistant, Department of Computer Science, Rutgers University

Fall 2026

HONORS AND AWARDS

Bronze Medal (team), ICPC China National Contest	2024
First Prize, Anhui Provincial Robot Competition (Programming Track)	2023
Second Prize, Anhui Provincial College Students' Mathematics Competition	2022
Outstanding Student Scholarship (Bronze), USTC	2022

SKILLS

Languages: Python, C++. **ML:** LoRA, DPO, RLHF, RAG, activation steering, membership inference. **Tools:** LaTeX, MATLAB, Mathematica.